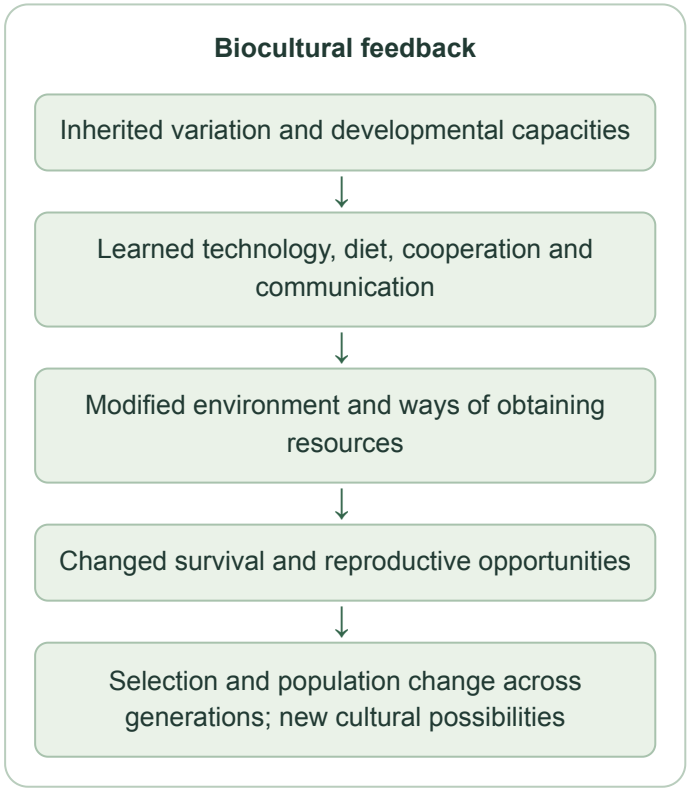
Human Evolution and Emergence of Man

1. The central idea: biocultural evolution

Human evolution is the interaction of inherited biological change and socially learned culture. Biological evolution changes populations across generations; learning changes individuals' behaviour and can alter the environment in which selection operates.

- **Sherwood Washburn:** biocultural feedback and the “new physical anthropology”; study adaptation and evolutionary processes rather than merely classifying human types.
- **Biology enables culture:** bipedal locomotion, dexterous hands, cognitive capacities, communication and prolonged learning/development.
- **Culture changes selection pressures:** tools, cooking, food production, settlement, disease exposure and social organisation reshape survival/reproduction.
- Individuals do not acquire a larger heritable brain by thinking harder. Mutations are not produced because an organism needs a trait; natural selection changes the representation of heritable variants.
- Humans share ancestry with other primates; **we did not descend from living chimpanzees**. Evolution is a branching history, not a ladder ending in humans.



Examples and limits

Interaction	Revision point
Dairying ↔ lactase persistence	Most infants produce lactase; many adults reduce its production. In some dairying populations, variants maintaining adult expression spread. Culture changed selection; milk drinking did not purposefully create the mutation
Cooking / food processing ↔ anatomy	Softer, more accessible food can change energetic and chewing demands; relevant to teeth, jaws and digestion. Exact timing and contribution to brain evolution remain debated
Tools ↔ hands, cognition and cooperation	Technology extends action and transmits knowledge; feedback involves anatomy, learning, environment and social behaviour, not a single “tool gene”
Settlement / agriculture ↔ disease	Population density, animal contact and diet change pathogen exposure and selection. Population resistance, individual immunity and public-health practices are distinct
Long childhood ↔ social learning	Prolonged dependence creates opportunities/costs of learning and care; cooperative caregiving matters, not necessarily one universal household form

Chronology trap: dairying, adult lactase-persistence selection and most farming/town examples belong mainly to the **Holocene**, not the Pleistocene. They remain useful examples of biocultural evolution.

2. Biology versus culture

Dimension	Biological evolution	Cultural change
Unit / information	Alleles, genotypes and inherited developmental systems	Learned knowledge, practices, values and technologies
Transmission	Primarily genetic parent–offspring transmission	Vertical, horizontal and oblique social learning
Mechanisms	Mutation, selection, drift and gene flow; recombination reshuffles variation	Innovation, teaching, imitation, diffusion and differential adoption
Pace	Can be rapid or slow; measured across generations	Often faster, but traditions can persist
Direction	No predetermined goal or automatic progress	Can be planned or unintended; not necessarily improvement
Individual versus population	Individuals develop/acclimatise; populations evolve	Individuals learn; practices spread/change in groups

Genotype ≠ phenotype. Phenotype reflects genotype, environment, development and their interaction. **Plasticity** is a capacity to develop differently under different conditions; not every plastic response is genetically inherited. Humans continue to evolve even when culture buffers environmental stress.

3. Pleistocene: setting of much human evolution

Pleistocene: c.2.58 million–11,700 years ago; repeated glacial/interglacial cycles, not one continuous worldwide ice cover. **Eocene: c.56–34 million years ago**, relevant to early primate diversification; it did not begin over 60 million years ago.

- Hominin evolution began before the Pleistocene. Bipedalism preceded the largest increases in brain size; australopiths were not a single species or a uniformly “cultureless” stage.
- Pleistocene changes included dispersal of **Homo**, tool diversification, changing diet, fire use, cooperation, shelters/clothing in some settings and eventually **Homo sapiens** dispersal from Africa.
- Fossils and archaeology permit inference, but precise origins of language, marriage, religion or a universal sex-based division of labour cannot be read directly from bones.

Archaeological association	Useful recall / qualification
Oldowan	Core-and-flake technology; associated with early Homo and other possible makers. “First tools = only Homo habilis” is too restrictive
Acheulean	Bifacial handaxes/cleavers; often associated with Homo erectus and later hominins; not every Homo erectus population used it
Middle Palaeolithic / Mousterian	Prepared-core/flake traditions; important Neanderthal associations, but not exclusive to one species. **Not Lower Palaeolithic**, as the source table implies
Upper Palaeolithic	Blades, diverse tools and prominent symbolic/artistic records, often associated with Homo sapiens
Mesolithic	Microliths, varied foraging and regional settlement changes
Neolithic	Farming/herding and more settled life in many regions; polished tools are a common association, not a universal definition
Bronze Age	Metallurgy, craft specialisation and urban/state formations in some regions; no global simultaneous sequence

Food, fire, brain and social life

- Foraging requires ecological memory, planning and learning. Hunting/scavenging could favour coordination and technological skill; meat was one resource among plant and animal foods.
- Fire offers warmth, predator deterrence, light, cooking and gathering places. Cooking can improve food accessibility and reduce chewing costs; adoption was uneven and its earliest timing is debated.
- Larger brains impose energetic/developmental costs; childbirth reflects pelvic dimensions, fetal size, metabolism and other factors. **“Hunting caused brain growth → women became immobile → compulsory male provisioning → marriage” is a historical model, not an established universal sequence.**
- Maternal care, male investment and help from other caregivers vary. Do not treat a particular gender division, incest rule or marriage form as a direct inevitable biological outcome.
- Brain regions in the chapter: **frontal—planning/executive and motor functions; parietal—somatosensory/spatial integration; temporal—auditory, memory and language-related functions; occipital—vision; cerebellum—coordination and learning**. These functions operate through networks, not isolated one-function boxes.

- Language supports cumulative knowledge and cooperation; **writing is a much later cultural invention**, not a prerequisite for human language. Social learning/tool use also occur in other animals; human cumulative culture has unusual scale and complexity.

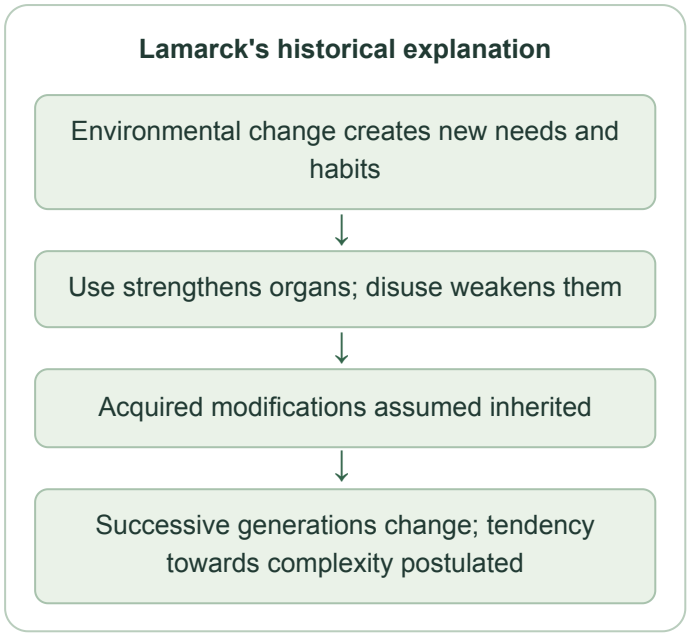
4. Before Darwin: organise ideas by contribution

Thinker / tradition	Main idea and evaluation
Mythological accounts	Indian **panchabhutas**, Dashavatara, Chinese elemental cosmologies, Egyptian/Babylonian creation narratives explain origins within cultural/religious systems. Dashavatara is not fossil evidence or a tested evolutionary mechanism
Thales / Anaximander	Water/primordial origins; Anaximander speculated on aquatic beginnings of humans. Important naturalistic speculation, not modern population genetics
Xenophanes	Fossil marine shells suggested past environmental change
Empedocles	Elements, combination and survival of suitable forms; an early precursor, not Darwinian selection in its modern sense
Aristotle, 384–322 BCE	Classification and ordered nature/scala naturae; not modern branching common descent. Humans are **viviparous**, not “ovoviviparous”
Lucretius	*De Rerum Natura*: naturalistic origins and change; separate poetic/philosophical speculation from empirical evolutionary theory
Special creation / fixity	Species created in present form; biblical chronologies included **James Ussher's 4004 BCE** calculation, published **1650**. Historical belief, not scientific age of Earth; “Jim Usher” is a source error
Hutton / Lyell	Deep geological time and processes operating over long periods. Hutton's **1785** presentation; Lyell's *Principles of Geology*, **1830–33**, helped supply Darwin's time framework
Maupertuis / Buffon	Heredity, variation and possible species change; precursors rather than a complete modern synthesis
Cuvier, 1769–1832	Comparative anatomy, extinction and **catastrophism**; opposed species transformation. Fossil replacement after catastrophes is not evolution by acquired characters
Erasmus Darwin	Suggested transformation, common ancestry and environmental adaptation; an intellectual precursor, not discoverer of the full Darwin–Wallace mechanism

Spontaneous generation proposed ordinary life emerging directly from non-living material. It is distinct from both evolutionary change among existing organisms and modern research into life's remote origin. The chapter's claim that Galileo was executed is false; he was tried and placed under house arrest.

5. Lamarckism: mechanism, importance and criticism

Jean-Baptiste Lamarck, 1744–1829; *Philosophie zoologique*, 1809. One of the earliest systematic transformist theories.



- Classic **giraffe** illustration: repeated stretching → longer neck acquired → descendants inherit it. This explains the historical claim, not the accepted genetic mechanism.
- **Value:** rejected simple species fixity, placed adaptation/environment in the explanation, developed invertebrate classification and a systematic account of transformation.
- **Context:** natural-history observation, Buffon/Erasmus Darwin, fossils/deep time, environmental change and Enlightenment-era ideas about transformation.
- Important works: **Flore française, 1778; Hydrogéologie, 1802; Philosophie zoologique, 1809**. Lamarck and **Treviranus** independently used “biology” in its modern disciplinary sense around **1802**; claims that Lamarck alone coined every relevant term are unreliable. **Eduard Suess**, not Lamarck, coined “biosphere”.
- **Criticism:** bodily use can enlarge a muscle within a lifetime, but does not establish inheritance of the acquired enlargement. No general mechanism links a need/use-induced bodily modification to the same directed inherited change; variation, selection and genetics offer stronger explanations.
- **Weismann's germ-plasm distinction:** soma and germ line differ; tail-cutting examples challenge inheritance of mutilation, though mutilation alone is not a complete test of every environmental-inheritance claim.
- **Epigenetic inheritance does not reinstate all Lamarckism.** Some regulatory marks persist under some circumstances, many reset; distinguish direct developmental exposure from inheritance across unexposed generations. No proof that desired beneficial changes are generally produced and inherited.

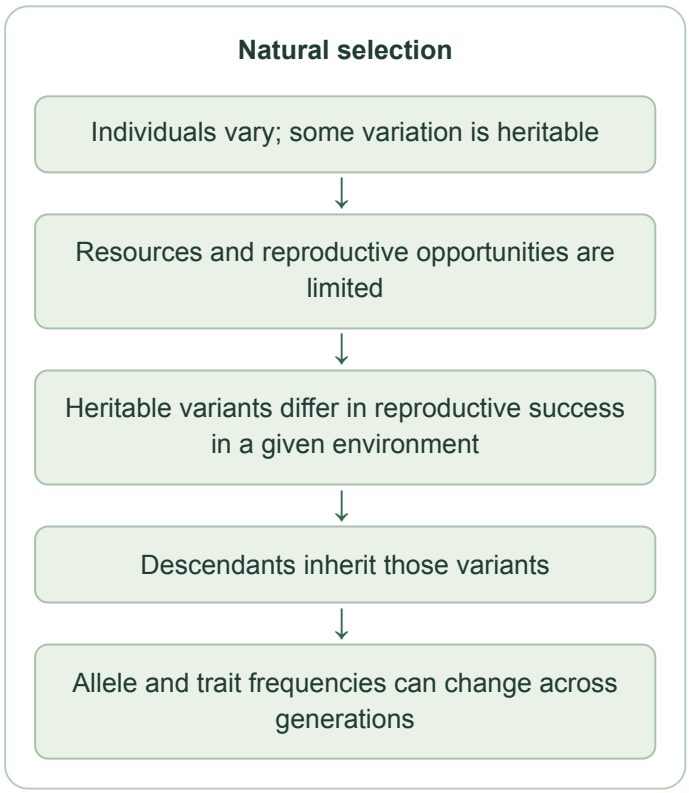
Neo-Lamarckism and experimental claims

Case	What to remember
Lysenko	Environmentally induced hereditary change claims received political backing in Soviet agriculture; suppression of genetics is a caution about evidence and political authority
McDougall's rats	Claimed progressively improved maze/escape learning across generations. Controls and replication matter; does not establish inheritance of learned knowledge
Sumner's temperature experiments	Body proportions changed with rearing conditions; offspring returning towards usual proportions illustrates difference between plasticity and inherited adaptation

Case	What to remember
Tower, Kammerer, Lindsey, Griffith claims	Historical experiments reported in the source on beetles, salamanders and other organisms; not accepted collectively as conclusive proof of acquired-character inheritance. Do not repeat precise outcomes as settled modern findings
Baldwin effect	Learning can improve survival and change selection on inherited learning capacity or associated traits. **The learned act itself need not become genetically inherited**
Cultural inheritance	Knowledge transmitted socially; real intergenerational transmission does not mean DNA carries the acquired idea

6. Darwin and Wallace: natural selection

Charles Darwin (1809–1882) and **Alfred Russel Wallace** independently developed natural-selection explanations; joint communication **1858**; Darwin's *On the Origin of Species*, **1859**.



- **Fitness = relative reproductive contribution**, not simply strength, longevity or moral superiority. Survival matters through reproduction; competition need not mean physical fighting.
- Mutation generates new alleles; **selection sorts existing heritable variation**. Variants are not generated because beneficial changes are needed.
- **Giraffe contrast:** selection on inherited variation in ancestral populations, rather than neck stretching being automatically inherited. A single simplified feeding story does not explain every aspect of giraffe evolution.
- **Population thinking:** individuals vary; populations change through time. **Common descent and branching diversification** replace fixed ideal types and a predetermined hierarchy.
- **Achievements:** adaptation, common ancestry and diversification within a naturalistic framework; evidence from fossils, anatomy, biogeography and breeding.

- **Limits of original Darwinism:** no adequate Mendelian account of heredity or molecular mutation, incomplete treatment of drift and mechanisms of reproductive isolation. Modern genetics strengthens and revises the framework; not every evolutionary change is adaptive.

7. Modern synthesis: Darwinism plus genetics

1930s–1940s synthesis: natural selection + Mendelian inheritance + population genetics + systematics + palaeontology. It was a collaborative development, not a single invention in 1937.

Contributor	Association
Fisher, Haldane, Sewall Wright	Mathematical population genetics; selection, variation and drift
Theodosius Dobzhansky	*Genetics and the Origin of Species*, **1937**
Ernst Mayr	Species, systematics, isolation and speciation
Julian Huxley	*Evolution: The Modern Synthesis*, **1942**
George Gaylord Simpson	Palaeontology and evolutionary tempo/pattern
G. Ledyard Stebbins	Plant evolution, variation and speciation

Core: produce/redistribute variation → selection and stochastic processes change populations → barriers to gene flow can permit divergence and speciation. **Microevolution and macroevolution are connected**, not "genetic change versus unrelated morphological change".

Mechanisms: do not confuse their jobs

Mechanism	Effect / example
Mutation	New DNA-sequence variation; point changes, insertions/deletions and larger changes. Effects can be beneficial, neutral or deleterious in context. Germ-line mutations can be transmitted in sexually reproducing humans; most somatic changes are not
Recombination	Crossing over, independent assortment and fertilisation reshuffle existing alleles; creates combinations, **does not by itself require change in allele frequency**
Gene flow	Alleles move between breeding populations through successful reproduction; movement without genetic contribution is not gene flow
Genetic drift	Random sampling changes allele frequencies; especially strong in small **effective** populations; can fix/lose alleles irrespective of advantage
Founder effect	New population founded by a small, non-representative sample; special drift setting, not a guarantee of greater diversity
Bottleneck	Sharp reduction of population size changes sampling/variation; population may regrow without recovering lost diversity
Natural selection	Non-random differences in reproductive success associated with heritable traits; context dependent

Mechanism	Effect / example
Hybridisation	Crossing differentiated populations/species can redistribute variation; may yield fertile, partly fertile or sterile offspring. Not simply any mating between unrelated people

Chromosomal changes: deletion, duplication, inversion, translocation, ring/isochromosome formation; number changes include **euploidy/polyploidy** (whole sets) and **aneuploidy** (individual chromosome gain/loss: monosomy, trisomy, nullisomy). Polyploidy can be important in plant speciation; do not assume a common successful human pathway.

Hardy–Weinberg: the reference model

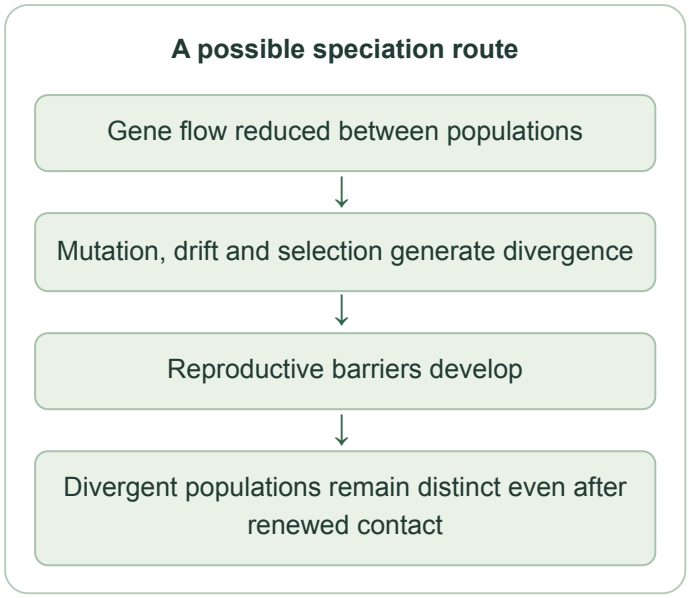
For a two-allele autosomal locus: **p + q = 1**; expected genotype frequencies **p^2 (AA) + 2pq (Aa) + q^2 (aa) = 1**.

- Ideal assumptions: **very large population, random mating, no mutation, no migration/gene flow, no selection**. Large size makes drift negligible. The source omits **no selection**; include it.
- Example: **p = 0.6, q = 0.4 → AA 0.36, Aa 0.48, aa 0.16**.
- It is a null/reference model, not a claim that every natural population is at equilibrium. **Non-random mating can alter genotype proportions without directly changing allele frequencies**; equilibrium deviations and allele-frequency evolution are not identical tests.

8. Isolation and speciation

Biological species concept: groups of actually/potentially interbreeding natural populations reproductively isolated from other such groups. Useful for many sexual organisms; difficult for fossils, asexual lineages and some hybridising groups.

Barrier / route	Example / significance
Geographical	River, island, mountain or distance reduces contact; **allopatric divergence** possible, but geography alone does not guarantee new species
Habitat/ecological	Different niches/host plants/microhabitats reduce encounters, even within the same broad region
Temporal	Different flowering/breeding times
Behavioural	Different courtship signals or mating calls
Mechanical / gametic	Mating or fertilisation incompatibility
Postzygotic	Reduced hybrid viability/fertility or later-generation breakdown



- Geographic, ecological and reproductive isolation are related concepts, not identical steps required in one fixed order. Speciation can also occur without a geographical barrier.
- **Genotype–environment interaction:** Richard **Lewontin** emphasised development and environment; no one-to-one rule that each gene produces one fixed trait.
- Different human populations are **not different biological species**. Human cultural isolation/endogamy can affect frequencies without implying a ladder of “races” or superiority.

9. Microevolution, macroevolution and later developments

Concept	Meaning / correction
Microevolution	Allele-frequency change within populations; investigated through genomes, pedigrees, blood groups and other variation. Not restricted to a fixed short duration
Macroevolution	Speciation, extinction, diversification and long-term patterns at/above species level; fossils, phylogenies, comparative and molecular data. Not only past events or only bones
Punctuated equilibrium—Eldredge and Gould, 1972	Long stasis interrupted by relatively rapid change associated with speciation on geological scales; not instantaneous transformation, and not a concept the original 1940s synthesis had already “settled”
Neutral theory—Motoo Kimura, 1968	Much molecular substitution explained by drift of selectively neutral/nearly neutral variants; does not deny selection on adaptations
Endosymbiosis—Lynn Margulis	Mitochondria/chloroplasts descend from bacterial endosymbionts; present organelles are **not free-living prokaryotic organisms**
Kin selection / inclusive fitness—W.D. Hamilton	Helping can spread when its relatedness-weighted benefits outweigh costs; **rB > C**. Not a conscious plan to preserve genes
Reciprocal altruism / multilevel selection	Other frameworks for cooperation; distinguish reciprocity from helping relatives. George C. Williams contributed to adaptation/selection debates; do not substitute him for Hamilton's key kin-selection formulation

Concept	Meaning / correction
Evo-devo	Evolution of developmental mechanisms/gene regulation shapes form; developmental constraints and plasticity matter

Mendel's work: presented **1865**, published **1866**. **Hugo de Vries' mutation theory, 1901–03**, emphasised discontinuous variation; modern mutation genetics is broader than his original saltationist theory.

10. Rules and patterns: learn the contrasts

Term	Meaning	Example / limit
Dollo's rule	Exact reconstruction of a complex ancestral condition through the same historical path is highly improbable	Whales returned to water without becoming ancestral gill-breathing fish. Not a ban on reverse mutation or reappearance of any trait
Cope's rule	Tendency towards increased body size in some lineages over time	Some mammal trends; not universal. Dwarfing, ecology and flight constraints supply counterexamples; “only unspecialised organisms survive” is false
Gause's principle	Competitors with identical limiting-resource requirements cannot stably coexist indefinitely under constant conditions	Competitive exclusion; partitioning niches can enable coexistence. **Not a definition of adaptive radiation**
Parallel evolution	Similar traits independently evolve from comparable ancestral starting points, often in related lineages	Repeated adaptations within related groups; precise distinction from convergence can depend on trait/developmental level
Convergence	Independent evolution of similar adaptive features in different lineages	Streamlined **shark, ichthyosaur, dolphin**; wings of birds/bats as flight structures. Their underlying forelimb bones are homologous
Adaptive radiation	Common ancestral lineage diversifies into forms occupying different niches	Darwin's finches; primate ecological diversification
Mosaic evolution	Different anatomical/functional systems evolve at different rates/times	Hominin bipedal adaptations preceded major brain expansion; not every trait changes together

Homology = shared ancestry of the feature; analogy = functional similarity independently acquired. Convergent wings can share homologous underlying limb bones. Palaemon is a **crustacean**, not an insect; omit it as a model fish-shaped vertebrate.

11. Adaptive radiation and primate examples

Ecological opportunity + heritable variation + differential adaptation + lineage separation → diversification. New habitats, altered resources or key innovations can create opportunities; no requirement that every ancestor was universally “unspecialised”. **Osborn** associated with early twentieth-century discussion; **Simpson** with adaptive zones and diversification.

Finch niches

- **Ground finches:** seed-crushing bills vary with seed size/hardness.
- **Warbler finches:** slender insect-foraging bills.
- **Vegetarian/tree finches:** feeding specialisations involving plant foods or insects.
- **Woodpecker finches:** extract invertebrates, including tool-assisted foraging.
- **Cactus finches:** elongated bills for cactus-associated resources.

Beak differences, behaviour and diet reflect ecological diversification; categories are a revision guide, not a fixed complete modern species count.

Primate niches and movement

Group / adaptation	Recall
Madagascar lemurs	Radiation into different dietary/activity/locomotor niches. **Aye-aye:** specialised incisors and thin finger for extractive foraging; **ring-tailed lemur:** mixed terrestrial/arboreal activity; **indri/sifaka:** vertical clinging and leaping
New World monkeys	Central/South America: **howlers—leaf-rich diet; spider monkeys—fruit-rich diet and suspensory movement**. More than just two families; do not memorise the source's incomplete family list
Tarsiers	**Southeast Asia**, large eyes and faunivory; not Madagascar lemurs
Chimpanzees / bonobos / gorillas	African apes; differing diets, locomotion and social patterns. Avoid a rigid “peaceful bonobo versus always violent chimpanzee” caricature
Orangutans	Southeast Asian great apes, flexible arboreal climbing/suspension; not African apes
Gibbons / siamangs	**Hylobatidae**, specialised **brachiation**; long arms and mobile shoulders
Humans	Apes with habitual terrestrial **bipedalism**; heel-to-toe plantigrade walking, **not toe-walking**

- Modern major primate division: **Strepsirrhini** and **Haplorhini**. Traditional “Prosimii/Anthropoidea” terminology is historically useful but does not capture tarsiers' placement well.
- Grasping hands/feet in many primates, nails/tactile pads, vision, flexible limbs and learning favour arboreal tasks. Locomotion varies: **quadrupedalism, clinging/leaping, suspension/brachiation, knuckle-walking and bipedalism**.
- Mammalian radiation illustrates modified homologous limbs: **arboreal climbers, bats in flight, cursorial runners, fossorial burrowers and aquatic swimmers**. Kangaroos are marsupials, so do not put them inside a list exclusively of placental mammals.
- **Plantigrade:** whole sole; **digitigrade:** digits; **unguligrade:** tips of hooved digits. Artiodactyl/perissodactyl refers broadly to even/odd-toed groups, not every member having exactly two/one toes.
- **Ancestral/derived** describes relationship to a specified ancestor, not inferior/superior. Living apes are independently evolved relatives, not unchanged human ancestors. No universal claim that the human brain is “twelve times any primate brain” is valid.

12. PYQ answer framework

For a theory: proponent/context → mechanism diagram → example → evidence and criticism → contribution to modern explanation. **For a process:** define → show genetic/population mechanism → distinguish a similar term → give a human/non-human example.

Recent mains questions

Year / marks	Question and answer cues
2021 · 15	**Explain the genetic mechanisms of micro and macro evolution.** Mutation/recombination, selection, drift, gene flow → isolation/speciation/extinction; connect levels rather than separating genes from morphology
2020 · 15	**Elucidate how Darwin and post-Darwin theories of evolution resulted in the development of Synthetic theory of evolution.** Darwin–Wallace → Mendelian inheritance → population genetics → Dobzhansky/Mayr/Huxley/Simpson/Stebbins → modern qualifications
2019 · 10 · about 150 words	**Implications of mutation in evolution.** New alleles and variability; effects context dependent; interaction with drift/selection; germ-line versus somatic; no need-directed mutation
2018 · 20	**Explain the biological changes that made human beings capable of making culture.** Bipedality, dexterity, cognition, communication and development; reciprocal feedback and mosaic timing
2016 · 15	**Elucidate the biological and cultural factors in human evolution.** Explain both and their interaction; tools, diet/fire, cooperation and later dairying; distinguish Pleistocene and Holocene examples
2015 · 15	**Critically examine Darwin's theory of evolution in understanding evolution.** Selection/common descent/population variation → examples → original heredity gaps → synthesis, drift and neutral change

Earlier mains themes

- **1997:** biological evolution of early man—sequence of traits and evidence; avoid ladder diagrams.
- **1994:** Darwinism versus synthetic theory—genetics, mechanisms of variation, drift and speciation.
- **1993 / 1992:** synthetic theory—integrate mechanisms and contributors; do not merely list names.
- **1992:** organic evolution and evidence for human evolution—fossils, anatomy, archaeology, molecular/biogeographic evidence.
- **1991:** organic evolution and its theories—historical Lamarck/Darwin alongside modern synthesis; assess evidence.
- **1990:** microevolution—population/allele-frequency definition and mechanisms.
- **1988:** organic evolutionary processes and speciation—variation, sorting and reproductive barriers.
- **1986:** biological evolution/new physical anthropology; Lamarckism versus Darwinism.
- **1985:** theory of organic evolution—definition, mechanism, evidence and modern integration.

The final source prompt **“Elucidate the theory of Evolution as put forward by Lamarck”** has **no year printed**. Treat as an undated revision prompt pending verification; it is excluded from year-based PYQ JSON. Earlier

source-supplied marks/word limits are retained in the common file with verification cautions in the review notes.

References for checking concepts

- Supplied chapter; Smithsonian: introduction to human evolution and human evolutionary evidence.
- UC Berkeley: mechanisms of evolution; Lamarck's life and theory; Berkeley population-genetics teaching notes.
- NHGRI: epigenomics and inheritance; Evershed and colleagues: dairying and lactase-persistence selection.
- International Commission on Stratigraphy: geological chart. Recent PYQ wording/marks checked against locally saved UPSC Anthropology Paper I papers.